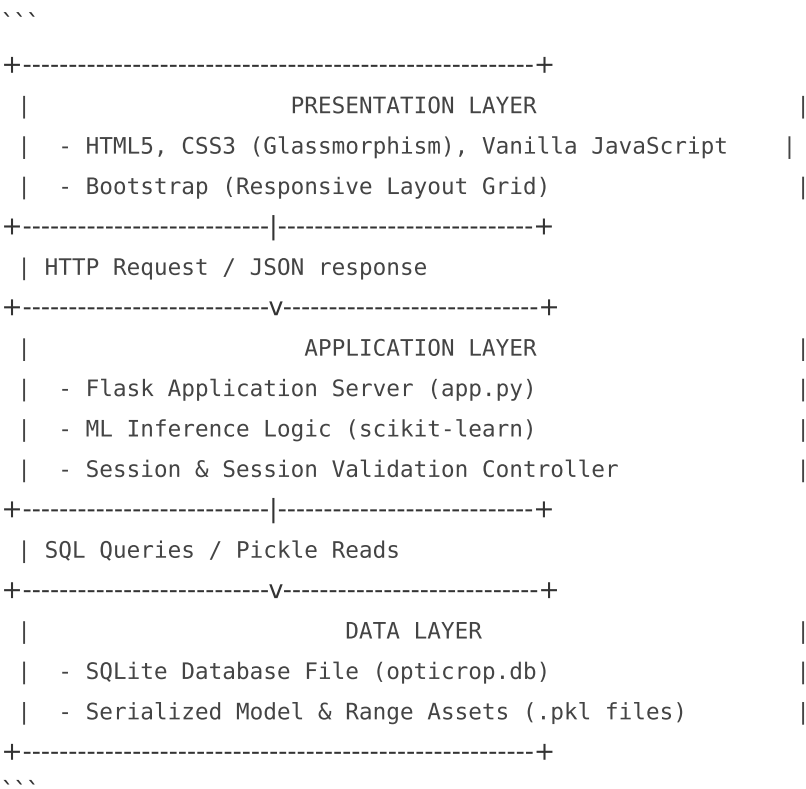


Solution Architecture

1. Three-Tier Architectural Layout

Opti-Crop is designed as a classic 3-tier web application, securing modularity, ease of testing, and clean separation of concerns.



2. Database Schema Relationships

The SQLite database (`opticrop.db`) maintains a structured schema with 7 interrelated tables to enable logging:

- `**`users`:**` Holds user profiles, hashed passwords, and access roles (Farmer vs Researcher).
- `**`soil_data`:**` Stores input soil properties, linked to the ``users`` table via foreign keys.
- `**`crops`:**` Stores optimal ph, season, water requirements, and crop categorization.

-
- **`datasets` & `ml\_models`:** Stores model tracking information (model accuracy and datasets used).
 - **`predictions`:** Links `soil\_data`, `crops`, and `ml\_models` with a timestamp and confidence score.
 - **`reports`:** Stores generated summaries and chemical/organic recommendation text.